CO4 AT - Question 1

0/1 Knapsack Optimization using Dynamic Programming

Aim:
To determine the optimal profit that can be obtained without exceeding the truck's carrying capacity using Dynamic Programming.

Problem Statement:
A warehouse manager must load items into a truck with a fixed carrying capacity. Each item has a specific weight and associated profit, and items cannot be split or reused. Determine the optimal combination of items to maximize profit without exceeding the weight limit using Dynamic Programming.

Input:
n = 3
Values = [60, 100, 120]
Weights = [10, 20, 30]
Capacity (W) = 50

Algorithm:
1. Create a DP table.
2. Initialize the table with zeros.
3. For each item, decide whether to include or exclude it.
4. Store the maximum profit in the DP table.
5. The last cell gives the optimal profit.

Expected Output:
Maximum Profit = 220

Dynamic Programming Concept:
The problem is divided into smaller subproblems, and their solutions are stored to avoid redundant calculations.

Time Complexity:
$O(n \times W)$

Space Complexity:
$O(n \times W)$

Result:
The maximum obtainable profit was found to be 220 without exceeding the truck capacity.

Conclusion:
Dynamic Programming efficiently solves the 0/1 Knapsack problem by reducing repeated computations and ensuring an optimal solution.

Output Screenshot:
(Insert terminal output screenshot here.)